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Constructing a Cohort: Experience with the French Elfe Project

French research into the outcomes, well-being and development of children is extensive but rather dispersed, carried out in demarcated disciplinary areas. However, an increasing number of researchers are seeking to pool their knowledge and are developing multidisciplinary studies to broaden the scope of exploration and dialogue in fields relating to childhood. The aim is to adopt a more comprehensive approach to the world of children and their development, drawing on varied areas of research. For example, collaboration between different branches of the social sciences provides a clearer picture of how cognitive skills are acquired by mobilizing knowledge of the social, family and cultural environment in which children grow up (Kaufman and Clément, 2003, 2007; Duru-Bellat and Fournier, 2007). In the field of health, epidemiologists more habitually adopt a multidisciplinary approach: the study of the causes and evolution of childhood diseases is not restricted to clinical diagnosis alone but also covers exogenous factors such as social background and family context.

Numerous statistical sources and data concerning childhood are available in France, but many gaps remain unfilled. First, these sources contain few

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direct observations of children's lives. To transcribe their life event histories, researchers generally rely on information from parents, relatives and other members of their familial or social environment. Historians usually approach childhood via the adult: "Research examines the action of adults on children to bring them out of childhood and into adult life. It does not examine children's own experience, and rarely their relations with each other" (Delalande, 2009). Furthermore, to overcome divergences between disciplines in the concepts used and in the interpretation of results, common analysis methods are needed. The data available to demographers, sociologists and health researchers come mainly from cross-sectional surveys and cannot be used for close examination of children's histories, their trajectories or the events that affect their lives.

A longitudinal approach is therefore needed to study child development. Following a cohort over time minimizes the memory bias inherent to all cross-sectional surveys (Riandey, 1995), provides a more finely grained picture of children's trajectories, identifies and orders the key events of childhood, moments of transition and changes of status (entry into and exit from poverty, transition from childhood to adolescence, etc.). The value of this survey method and the opportunities it offers in both health research (epidemiology, aetiology, well-being) and social sciences has contributed to the development of the "life course approach" (Kuh and Ben-Shlomo, 1997; Elder, 1998; Heinz, 1991; Bynner, 2001), a dynamic method for identifying and observing the long-term effects of physical and social exposures and their interactions during childhood, adolescence and adult life. A cohort study provides a means to monitor these various factors over time, to identify certain pathologies, for example, at birth and to track the various stages in child development (Grandjean, 2008).

Multidisciplinary projects with cohorts of children monitored from birth (or soon after) to adulthood have existed in various countries for many years (Pirus and Leridon, 2010), but not in France until now. The Elfe project (Étude longitudinale française depuis l'enfance) launched in 2011 aims to close this gap by monitoring 20,000 children from birth to adulthood via a multidisciplinary approach. The project has been designed to provide a better understanding of child development, particularly with respect to their physical and social environment. The children will be observed at regular intervals using various data collection methods, such as in-home surveys, diaries and telephone interviews, to evaluate and accurately measure the factors – familial, social, environmental, health, medical or nutritional – that affect their development and to observe the impact of situations experienced through childhood to adulthood.

This article describes the development of the Elfe project (www.elfe-france.fr). Section I presents its origins, its specific features and major scientific objectives. Section II details the methodological aspects of the study and the first lessons learnt from the pilot surveys carried out in 2007 and 2008.

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I. Designing the Elfe project

To construct the Elfe project, the design team drew upon the experience with large cohorts acquired by other countries (Pirus and Leridon, 2010) and upon the findings of smaller research initiatives in France. Until now, no large-scale multidisciplinary longitudinal study of child development had been carried out in the country.

1. Origins of the project

For researchers, the interest of such a study was obvious. But demand also came in a more specific shape from various government and statistical bodies, such as the Ministry of Health, the Ministry of Social Affairs, the council for employment, income and social cohesion (CERC*)(1) and the national council for statistical information (CNIS*) following the report of the mission on panels – organized jointly with the French statistical office (INSEE*) – presented to the social statistics joint working group in September 2004 (Chaleix and Lollivier, 2005) and adopted by resolution of the CNIS on 11 February 2005. The report on children's and young people's health from the higher public health committee (Haut comité de la santé publique), written for the national conference on health 1997,(2) proposed "longitudinal studies from early childhood in order to track difficulties in fundamental skills acquisition, accidents and violence, to analyse risk behaviour and evaluate the impact of measures taken" and "multidisciplinary research involving the social, clinical and epidemiological sciences to identify the origins of high-risk situations and behaviours". When it presented its draft medium-term strategy to CNIS in 2003, the department for research, studies and statistical evaluation (DREES*) pointed out that:

> Recent analyses of poverty and the impact of social transfers have shown the value of considering the child as a basic statistical unit. This implies, where appropriate, adapting household surveys to permit this type of approach – as is already the case in English-speaking countries – and undertaking specific surveys relating to the "child" in the household. In this context, it would be particularly useful to bring together a wide-ranging set of partners to establish and monitor a cohort of children to be tracked from the earliest age with a view to evaluating the impact of situations experienced during childhood on individuals' subsequent social integration and mobility.

On this basis, discussions began in the second half of 2002. An initial preliminary draft for a "national child cohort" (Cohorte nationale d'enfants, CNE) was circulated first within the French National Institute for Demographic Studies (INED*) and then within the other bodies mentioned above (CERC, DREES, INSEE). Meanwhile, the management of the National Institute for Health and Medical research (INSERM*) had also discussed the cohort issue

(1) Full titles and translations of acronyms marked with an asterisk are given in Appendix 4.

(2) Available in French at http://www.hcsp.fr/explore.cgi/ouvrage?ae=ouvrage&clef=11&menu=090471.

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and the CNE project was proposed, with greater emphasis on health aspects. But these points were considerably modified when another project emerged.

In June 2004, the French government adopted a national plan for health and the environment to increase the resources available for reducing adverse environmental impacts on health. The plan included in its priority recommendations an "epidemiological study of children in line with the US National Children's Study", recommending "the monitoring of a cohort of children in France, including objectives of surveillance and research, to be part of a national partnership built around The French Institute for Public Health Surveillance (InVS*) and INSERM" and suggesting a sample size of between 10,000 and 20,000 children. Under the auspices of the InVS, discussions were held from July 2004 to June 2005 between the various possible partners, including potentially interested research teams. It rapidly became clear that this second project (known as EFESE) and the CNE project overlapped on too many respects to remain separate. In June 2005, therefore, it was decided to merge the two under the acronym Elfe (Étude longitudinale française depuis l'enfance). The main partners in the new project were INED, INSERM, InVS, the department of health (DGS*) and INSEE, later joined by DREES, the evaluation and perspective department of the Ministry of Education (DEP*) and the national family benefits fund (CNAF*). Together, these bodies formed a scientific interest grouping (*groupement d'intérêt scientifique*, GIS) in March 2006 and decided to base the project management team at INED. In March 2010, the GIS was replaced by a dedicated INED-INSERM joint research unit, with INED still providing the logistical base (see Appendix 3 for the list of participants in project design).

At the time Elfe was launched, the French research funding system did not make standard provision for this type of necessarily long-term project. Furthermore, the health component in Elfe involves the collection of biological samples, which considerably increases the total costs. The usual research funding channels (French National Research Agency (ANR*), public bodies and foundations) and the bodies co-ordinating the project (INSERM, INED, INSEE, etc.) could not provide the necessary funding beyond the project preparation period and pilot stages (from 2005 to 2009). The decision to support the project from public funds was finally taken in mid-2009, at an interministerial meeting that reorganized its governance (as the GIS was coming to an end) and earmarked funding for the first two years – by far the most expensive phase – from the budgets of three ministries: those responsible for research, health and the environment. This funding was confirmed at the end of 2009 as part of a large-scale research facility (*très grande infrastructure de recherche*) created by the Ministry of Research to support the construction of major health cohorts, including Elfe. At the time of writing, funding is only confirmed for the earliest stages of data collection (the most expensive) in maternity units and at age two months.

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The absence of guaranteed funding makes it difficult to maintain the mobilization of the many research teams involved in the process. This posed a serious challenge.

2. Project features

A deliberately multidisciplinary approach

These various factors led to the idea of developing a common tool: a panel of children for multidisciplinary research bringing together research teams from a variety of backgrounds. Two successive calls for proposals were launched among researchers in the social sciences and health in May 2005 and June 2006, along the following general lines: the cohort would be formed on the basis of a (nationally) representative sample of births of some 20,000 children, with data collection beginning in maternity units and continuing mainly via in-home or telephone interviews roughly every year until at least age 6. The cohort would be "general" (i.e. not restricted to any particular theme) and the observation tools would be specifically suited to this type of cohort. The respondents to the call for proposals were required to show in what way longitudinal observation was essential to their research theme. After excluding all unsuitable proposals, those remaining (approximately 90, from some 60 teams) were grouped into themes, so that each thematic group proposed a coherent and viable set of observations and analyses. The requests from various groups were then merged to construct the final observation instruments. The idea was to pool analytical tools and construct indicators that could be used in the various fields of research concerned by the cohort, and then to link up the phenomena under study with a view to broadening interdisciplinary scope.

For example, the study of families in poverty involves analysing their living conditions on the basis of information about their housing, financial resources and employment situation. But it may also be linked to their quality of life in terms of access to health care (consumption of care, access to dental care, for example), so the two themes must be developed in close consultation. Some themes are cross-cutting by their very nature, such as children's nutrition, which may be observed both via the transmission of eating habits within the family and the role of dietary socialization, and via the impact of nutritional inputs on child growth. The system for monitoring food intake (diary) used in Elfe is a good example of a multi-disciplinary tool since it is useful in both research areas and also ensures regular contact with the families during the children's first year.

Another advantage of this coordinated multidisciplinary strategy is the capacity to transpose the explanatory approach of one specific research area to another. An independent variable in one field may become a dependent variable in another. For example, in epidemiology, the onset of a pathology (respiratory disease, disability) developed by a child is a variable that can be explained by various risk factors (birth weight, congenital malformation, environmental factors, etc.), while in sociology it can be introduced as an

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explanatory variable for analysing the children's motor development and socialization; it is well known, for example, that a child's social integration may be restricted by a disability.

In more general terms, many social, demographic and economic variables will be used in epidemiological approaches. Conversely, data on child health can be included in certain areas of social science research.

Difficulties in collaborative research design

The idea of associating research teams from various disciplines was an innovative one, but the difficulties in achieving this goal were underestimated at the outset. The teams found it hard to join forces in a common project, and sometimes tended to defend their own interests too exclusively. The place assigned to each theme was not defined in advance, but some themes emerged as more important than others as the project took shape, as can be seen in the "skeleton" of the survey (see below for details of the survey "backbone"). The first research teams began working on the Elfe project in September 2005, but the main points of the methodological framework were only settled and decided upon two years later, and funding was not finally guaranteed until 2009. In practice, it was necessary to keep the various requests for observations within reasonable bounds, to define the questionnaire time allocated to each team, and to identify the points at which each theme fitted best into the monitoring timetable, taking account of the children's age and the importance of each discipline. For example, the first assessment of vocabulary acquired will take place at age 2, and the British Ability Scale (BAS) will be used to measured the children's cognitive development at age 3, so sufficient time must be made available in the questionnaires planned for those two ages. Later, at ages 6-7 (entry into primary school), collecting information on school performance and attitudes to school and educational learning will take priority.

Consensus had to be achieved in data collection methods to satisfy the various disciplines and research projects. But each discipline has its own priorities. Epidemiology researchers, for example, wish to use internationally validated question modules (important for having articles published in major journals) and to confirm as rigorously as possible the pathologies detected in the child, and this may require time-consuming extra examinations. Sociologists, meanwhile, want to carry out qualitative surveys on specific population groups in addition to the planned schedule (as is the case for the Growing Up in Ireland cohort; see Pirus and Leridon, 2010). However, whether medical or sociological, the surveys of sub-groups carried out under specific criteria are likely to be poorly controlled and may threaten the monitoring process as a whole, with no real input to the multidisciplinary approach.

The researchers responsible for the various themes (twelve initially, rising to more than fifteen) worked with one of the sector coordinators recruited

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nationally (in social sciences, health and health-environment), who made the necessary choices. These coordinators also ensured that certain themes were not "forgotten" if there was no spontaneous proposal from the research teams. For example, a group had to be formed to cover the "physical growth" aspects neglected in the proposals received. Discussions were also held at meetings of the "scientific project group" comprising the national coordinating team and the theme coordinators. The scientific council finally conducted an overall assessment of the projects in early 2010. It was also agreed that no team would have exclusive property of any part of the data or of any specially constructed indicators. For example, if an economic insecurity indicator is designed by a particular team, it may be used as an explanatory variable by the other teams. A charter will specify these rules of access and use.

Making good use of the life course approach

The Elfe cohort also corresponds to the life course approach developed over the last twenty years to examine individual trajectories from a multidisciplinary angle. In epidemiology, the life course approach examines the biological, behavioural and psychosocial processes that come into play over a lifetime and influence an individual's state of health (Kuh and Ben-Shlomo, 1997; Kuh and Hardy, 2002). It is a dynamic approach to development that identifies and observes the long-term effects of physical and social exposures and their interactions before birth, in childhood, adolescence and adulthood. Cohorts enable us to accurately record the timing of these various factors, to identify, for example, certain pathologies at birth and the various stages of their evolution. The positive and negative events in a life cycle evolve and interact. Ultimately they may have a beneficial or detrimental effect on an individual's health. In health (and also in epidemiology and aetiology), there are basically two models in the life course approach: one examines the links between an exposure, an event observed at one point in the life-cycle, and the likelihood of observable consequences occurring later in life, and the other measures the effect on health of multiple or recurring experiences during life; for example, continuous exposure to poverty that may cause health problems. In sociology, the life course approach emerged in the 1980s and is defined as a set of rules that organize the basic dimensions of the individual's social life (Kohli, 1985). In recent years, adjustments have been made to the "traditional" life course, represented in a linear fashion by three periods in a person's life: education, employment/family life, and retirement. This pattern has been somewhat destructured in response to changes in the labour market, and in the lives of young people and families (Pirus and Leridon, 2010). According to Guillaume (2005), the life course is to be observed as a "social institution both socially organized and constructed by individuals", with an individual level (individuals' experiences) and a structural level (model with a system of norms, a set of collective representations, etc.). Bourdieu (1986) also points out that in surveys based on life histories, respondents are not

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"neutral" when they talk about their lives, and they "act" in line with certain norms and social representations.

Frequent, varied observations

The first stages of child development take place very quickly and their varied aspects cannot be accurately described on the basis of information collected only every few years. Ideally, monitoring should begin during pregnancy, and this option was discussed at length at meetings in 2005-2006. It was concluded, however, that France's statistical and healthcare systems would be unable to construct a nationally representative sample of pregnant women who could be questioned at a sufficiently early stage in their pregnancy. Observation would begin in maternity units in the days immediately following birth, on the model of the national perinatal surveys (Enquêtes nationales périnatales, ENP) already carried out in France. The information gathered by Elfe in maternity units would thus constitute a new national perinatal survey, since the Elfe sample was constructed using the same method but on a larger scale. Finally the DGS considered that the start date of Elfe data collection would be too distant from the last ENP, carried out in 2003, so the two operations were separated (the new ENP was held in March 2010). The ENP protocol comprises an interview with the mother and the collection of medical information on her pregnancy; to these were added a self-completion questionnaire on diet and various exposures during pregnancy, and the collection of biological samples (described in detail below). We scheduled an in-home or telephone interview of 50-60 minutes with the mother 6 to 8 weeks after the birth, and an interview of 20-30 minutes with the father on the same date. Elfe will thus gather a considerable range of information on the pregnancy, the birth and the baby's first two months never previously recorded on such a scale for a nationally representative sample.

Two different sources and procedures will be used to obtain information about the children. The direct method involves conducting regular surveys at particular ages, using procedures such as in-home interviews, telephone interviews and biological sampling. The indirect method is used to collect data continuously from diaries (nutrition, for example) or from external sources such as the social security system for access to healthcare and medical prescriptions, and geocoded information about physical or chemical exposures. The provisional timetable for the various regular and continuous data collection procedures is given in Appendix 2.

The direct surveys, based on individual questionnaires administered to the parents and later to the children,[3] will collect socio-demographic information

(3) In cohort studies in other countries, the child is usually questioned directly from the age of 7 or so, but this does not mean that the questioning of the parents stops. The Millennium Cohort Study in the UK asks children of 7 to complete a questionnaire about their leisure activities, friends, school and feelings. In the Growing Up in Ireland cohort, children are interviewed directly at age 9. A short questionnaire covers life in their family, with friends, at school, activities, leisure and food preferences.

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about the children's family environment, psychomotor development and state of health, nutrition and exposure to environmental factors. Passive monitoring will provide information on more targeted themes such as environmental pollution (placing of sensors in the children's home and outside air measurements), consumption of healthcare (from data on health expenses in social security files) and family benefits (data from the CNAF). These last points will naturally require specific consent from the families.

The biological samples to be taken and health examinations to be made under Elfe are as yet only clearly defined for the first year of life. Samples of umbilical cord blood, mother's urine, venous blood, hair and mother's milk (if the mother breastfeeds) will be taken in the maternity unit; these samples will provide information about exposure to certain pollutants and nutrition during pregnancy. Other biological collections are scheduled at ages 3 and 6.

Once the children enter primary school, the Elfe cohort sample could be used for various school surveys conducted by the Ministry of Education. At present, we cannot confirm the possibility of collaboration in this area, for two main reasons. First, the ministry surveys generally use cluster sampling, i.e. whole classes drawn from a sample of schools, which clearly do not overlap with the Elfe sampling method. Second, civil society is often wary and may strongly object to collection of data that include the socio-demographic characteristics of family members, and children's academic skills and health status. When a "pupil base" file was created in 2007 to help with the administrative and educational management of pre-school and primary pupils, some teachers and parents strongly objected to head teachers making entries in the database, since the records contained personal information on children and their families. In response to this opposition, the Minister of Education decided in 2008 to remove data on nationality, year of arrival in France, language learning and culture of origin. However, the assessment of educational achievement is a key component of Elfe, and as Alain Desrosières (2010) rightly pointed out on the first World Statistics Day,

> […] for years voices were raised to criticize the recording of information on the occupations of parents of public and private primary school children, for fear that these children might be labelled, even unintentionally. And yet without this information, the sociology of inequalities at school would have been impossible. Similar arguments are made about the proliferation of computerized records. This does not mean that these criticisms must be ignored. But it should be possible to express them in places where the needs and dangers can be openly discussed. One such place is the CNIS, which brings together representatives of public statistical services and 'social partners', trade unions and voluntary associations […].

Other ways of collecting information on children's educational trajectories will need to be considered.

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The place of fathers

The second innovative feature is the involvement of the children's fathers in the Elfe cohort. Few cohort studies in other countries monitor children from the father's point of view, and most collect data on the children by interviewing the mothers. The place and function of fathers in family life have greatly changed in recent decades and this question is of considerable interest to researchers in family sociology. French society has shifted from a basically patriarchal model to a system in which new roles are attributed to each parent, implying an active (in terms of time spent) and affective presence of the father with the child (Théry, 1998). For the Elfe project, it was seen as crucial to fully involve the father and contact him as soon as possible, starting with the interview at 6-8 weeks, (under the chosen sampling method, initial contact is made with the mother in the maternity unit). The father will then be solicited regularly, even if the couple separates. Few studies in other countries question the father, or ask similar questions to both parents. Only the UK Millennium Cohort Study questions both parents on identical themes: during the first wave, the themes common to both parents in face-to-face interviews concerned their "roles" and attitudes as parents, their educational level, labour market status and state of health. Each parent was also asked to fill in a self-completion questionnaire concerning depression and opinions on various social topics (marriage, sexuality, etc.). The 2007 report on the Canadian NLSCY study recommends the full involvement of fathers in the next child cohort project.

In parallel to changes in the place of the father, new family structures have emerged (reconstituted families, lone-parent families, etc.) and new forms of parenthood, with an increasingly sharp distinction between biological and social parenthood (Godelier, 2004; UNAF, 2007). A parent is no longer simply a person who has recognized a child and/or is their biological progenitor, but also a person who participates in their education, development and sociability. The term "social parent" is used for an adult who is neither the biological nor legal parent but does play a parental role in the child's life. In same-sex families, this is often the partner of the biological or adoptive mother or father. These parents are not recognized in French law: to date they have neither rights nor duties with respect to the child, nor can the child claim any rights from their social parent(s). For the Elfe cohort, if the parental couple breaks up, the "primary carer" will be questioned, namely the one with whom the child usually lives (in cases of alternating custody, there will be two primary caregivers), and, as far as possible, the one who is no longer cohabiting with the child. It is also intended, if a new couple is formed, to question the primary caregiver's new partner, defined as a social parent, even if he/she is not the child's legal parent. When the social parent replaces the legal parent in the child's home, the same modes of observation and monitoring will be implemented as would have been the case for the legal parent.

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Links with cohorts in other countries

Naturally, a project of this sort inevitably draws upon the experience of other countries. Early contacts (2004-2005) were made with researchers in the UK (for existing cohorts, managed by the Centre for Longitudinal Studies), Canada (federal and Quebec surveys) and the US (mainly for a project under preparation, the National Children's Study). The main child cohorts are presented in another article (Pirus and Leridon, 2010). Meanwhile, a project for a European Child Cohort Network (Eucconet) was approved by the European Science Foundation (ESF) in late 2007. The multidisciplinary dimension of this project, associating health and social science cohorts for the first time, required twofold approval by the ESF. Eucconet is a European network based at INED in Paris covering ten countries. Its role is to examine the development of longitudinal analysis tools and to take advantage of the knowledge and experience of the various participating countries. More specialized topics are discussed in specific seminars, such as the father's involvement in child cohorts, methods for maintaining family participation over time, and cohort database management (www.eucconet.com).

3. Main scientific objectives

After discussions among researchers and the members of the Elfe scientific council, it was decided to organize Elfe around a central backbone in order to safeguard the project's multidisciplinary dimension and longitudinal nature, and to consolidate the various themes already included in its scientific programme (Appendix 1.A). When the project was first set up, and further to the calls for proposals, 14 themes were initially defined: 3 in social sciences, 4 in health, 3 in environmental health and 4 cutting across the three disciplines.[4] This central backbone was then organized chronologically according to the children's age and structured in a long-term perspective. Consequently, some themes will be studied more intensively during the first years of the children's lives, while others will be studied later (for example, sexuality at puberty).

The topics relating to living conditions and the environment will focus on:

- antenatal environment, as measured by exposures and biological markers sampled at birth;
- children's family, economic, cultural and social environment: changes in family structure (breakdown, reconstituted family, deaths, etc.), forms of custody, intergenerational relations, parents' income and occupations, housing conditions, characteristics of the neighbourhood (distance from amenities, industrial sites, etc.);
- parents' and children's eating habits (duration of breastfeeding, transition to solid food, etc.), physical activity, learning within the family, cultural

(4) On 31 December 2010, the list comprised 19 thematic groups (Appendix 1B).

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activities, exposure to media and new technologies (computers, video games, etc.);
- physical and chemical environment: exposure to "emerging" pollutants, air pollution, dust, etc.

Regular cohort monitoring will enable researchers to study the impact of these various factors and their interactions on the main aspects of child development:

- physical (growth, puberty) and psychomotor development (learning to walk, fine motor skills, understanding of the environment, first words and development of vocabulary and grammar);
- socialization process and educational trajectory (assimilation of social values and norms, school learning, social and occupational integration, etc.);
- health: the most frequent pathologies of childhood and adolescence (overweight, obesity, asthma and allergies, accidents and injuries, mental and behavioural disorders), risk markers for certain adult diseases, and healthcare consumption.

A range of early exposures and health markers will be assessed using biological sample collections. Samples will be taken to measure toxic exposures (umbilical cord blood, mother's blood and urine, meconium, mother's hair at birth); nutritional exposures (cord blood, colostrum, mother's blood); metabolic markers, inflammation, oxidative stress of mother and fetus (cord blood, mother's blood and urine); infectious contacts (cord and mother's blood, child's first stools); fetal RNA (cord blood); epigenetic phenomena (DNA taken from white blood cells of mother and fetus or from cord cells).

The Elfe project should permit major scientific advances in many areas. It aims to:

- identify, among the environmental pollutants currently causing concern (flame retardants, phthalates, bisphenol A, pesticides, heavy metals, indoor and outdoor air pollutants in general), those which present a short- or long-term risk for vulnerable persons (pregnant women, unborn babies, young children) at the concentrations currently observed in France. The study data will also provide information on ways of reducing this risk;
- describe practices of infant feeding and nutrition for which no national information is available in France (duration of breastfeeding, introduction of solid foods, etc.), their social and cultural determinants, their relationship with subsequent eating habits and health (obesity, allergies, etc.). The aim is to find answers to certain key questions that remain unanswered today. For example, to reduce the frequency of food allergies, should infants be exposed to food allergens or, on the contrary, should they avoid exposure? Does the introduction of solid foods before 6 months encourage a more varied diet in later life, and can it help to prevent obesity?

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- study social inequalities in health affecting children: even at birth, a social gradient for prematurity, intra-uterine growth retardation and big baby syndrome (macrosomia) is observed. How do these inequalities evolve over time, particularly in relation to differences in healthcare use? At what age does the social gradient appear for child pathologies such as obesity? Could these social inequalities be reduced by providing early support to young women entering their reproductive lives, to pregnant women and new parents?
- understand more fully the different factors (individual characteristics, physical environment, places of socialization) which interact at a very early age to influence progress through school and, in particular, the development of cognitive, language and socio-affective skills. Can these data provide new insights for reducing the school failure rate in France?
- analyse the effects of changing conjugal behaviours and the growing number of lone-parent or reconstituted families on children's social, health and educational development. These observations will be tied in, for example, to the way skills are learnt within the family or to the circumstances whereby families fall into poverty;
- draw up new reference curves for physical growth; the standard curves currently in use do not take account of recent changes in lifestyles, eating habits, etc.

II. Methodology

1. Sample design

The principle of a survey beginning at birth rather than during pregnancy was adopted in 2005 (Salines, 2005). The idea was to draw a sample from all the mothers giving birth in metropolitan France (mainland France and Corsica) on the days chosen by INSEE for the new permanent demographic sample (Échantillon démographique permanent, EDP), namely the first four days of each quarter (starting on 2 January rather than 1 January for the first quarter) (Couet, 2006). This ensured that the sample would be adequately representative and would take account of seasonal variations while enabling Elfe to make use of the information gathered for the EDP: first, data from birth records and successive censuses, and later, other data that INSEE intends to add to the EDP. The fact that Elfe is a sub-sample of the EDP will enable us to monitor the representativeness of the sample as attrition increases. This 1-in-23 sampling base (16 days out of 365) should represent some 35,000 births and will be used to form the target sample of 20,000 births.

Collecting the samples in the maternity units introduced a further constraint, namely the twice-daily collection and transport of samples to centres able to process and freeze them. After contacts with the French blood management authority (EFS*), the only institution capable of performing this task at national

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level, it was decided to limit sample collection to a total of 10,000 births (shown by statistical power calculation to be sufficient for most of the processing envisaged), to reduce the number of collection points and to extend the recruitment periods. This last constraint turned out to be an advantage, since the additional time available will enable the teams in place to deal with any technical hitches. Consequently there was a partial move away from the EDP: only half the children will come from the EDP sample. The final schedule is as follows:

1st period: children born on 28, 29 March, 1, 2, 3, 4 April 2011

2nd period: children born on 27, 28 June, 1, 2, 3, 4 July 2011

3rd period: children born on 27, 28 September, 1, 2, 3, 4 October 2011

4th period: children born on 12, 13, 16, 17, 19, 20 December 2011

A pause of 48 hours is scheduled after the first two days' recruitment in each period to give the interviewers time to organize their workload.

There are 539 public and private maternity units in metropolitan France. To recruit 20,000 children during the selected periods (24 days in all) while maintaining national coverage, maternity units were drawn at random after stratification by size (number of births expected). Five strata containing an equal number of maternity units were established, and the draw over-represents large units (random draw proportional to the number of births expected in each stratum, taking 100% of the stratum of largest units). Within each stratum, implicit stratification[(5)] by region, technical facilities available in maternity unit and public/private status ensured a fair distribution on the basis of these criteria. For the descriptive analyses of the results, the sample will naturally need to be weighted to correct for the over-representation of large maternity units.

In all, a sample of 344 maternity units was drawn, with a potential of 40,000 births during the Elfe recruitment days. Reckoning with a failure rate of 10% at maternity unit level, and 45% at individual level (percentages observed in pilot studies), this number of maternity units should be sufficient to recruit 20,000 children.

For reasons of cost and organization, biological samples will be taken from a sub-sample of Elfe maternity units up to a total of 10,000 births. The maternity units participating in the biological collection will be those in the initial sample located less than 100 km from a sample processing centre.

All the births occurring during these periods will be included, except for births before 33 weeks of amenorrhoea (around 1%), multiple births of more than two children (around 0.03%), and children born to underage parents or parents unable to give informed consent.

(5) Implicit stratification involves ranking the various units by each variable to be considered and then carrying out systematic sampling (at sampling ratio T defined in the overall sampling design, i.e. one in T units from the full list).

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Very preterm babies (less than 33 weeks gestation) will be followed by the Epipage 2 survey which covers all the Elfe recruitment periods (http://www.ffrsp.fr/8.html). The Elfe team is working in close collaboration with the leaders of this project.

2. Lessons learnt from the 2007 and 2008 pilot studies

Two pilot samples

To test the feasibility and acceptability of the Elfe study, two pilot studies, each with two phases, were held in 2007: the first in April in Burgundy and Picardy, and the second in October in four *départements* of the Rhône-Alpes region and in the Seine-Saint-Denis *département*. Unlike the April pilot study, the October study included biological samples taken in the maternity units. During this exploratory stage, we deliberately chose to hold the study in Seine-Saint-Denis, where economically and socially vulnerable and non-French population groups are over-represented. The first phase took place in the maternity units, when the children were born, and the second in the children's homes, at age two months, in the form of a face-to-face interview with the mother and a telephone interview with the father. These sampling methods provided a "full-scale" test of the chosen collection procedures, since they were used for all births over four consecutive days in these ten *départements*.

Maternity units vary enormously in complexity, size, level[6] and composition of the medical staff. Preliminary work is therefore necessary to organize and coordinate the maternity unit survey phase in cooperation with hospital personnel. This is a relatively complex operation for two reasons. First, this is precisely the moment when families are recruited, and experience with other cohorts has shown that it is often at this stage that the greatest difficulties arise. Second, the maternity unit survey involves a series of tasks to be carried out in quick succession among women who have just given birth. The main stages are: presenting the Elfe cohort to the new mothers, asking for a long-term commitment, gathering information from their medical records, holding a 20-minute face-to-face survey, taking biological samples and handing out a self-completion questionnaire about their eating habits and lifestyle during pregnancy. The time available for the survey is the duration of the women's stay in the maternity unit, which is less than four days on average.

The participation rate (for the two pilot surveys), calculated as the ratio of families who agreed to participate to the total number of births in the maternity units during that period, was 55% (Table 1), with large variations across maternity units and *départements*. But this rate only reflects an average for an exploratory phase during which the major difficulties likely to arise in the full-scale study could be observed and analysed. The participation rate was

(6) Maternity units in France are classified in three levels: uncomplicated pregnancies are sent to Level 1 units, Level 2 units include a neonatology unit, and Level 3 units have a neonatal intensive-care unit. The level is also generally linked to the size of the maternity unit (number of births per year).

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Table 1. Percentage of mothers interviewed in the first 3 phases of the Elfe pilot study

	Child's age			
	Birth	2 months	1 year	2 years
Number of eligible births	834[a]	450	419	390
Number of interviews in each wave	455	384	343	318
Percentage interviewed in each wave	54.5	85.3	81.9	81.5
Percentage interviewed out of total eligible births in maternity units	54.5	46.0	41.1	38.1
Percentage interviewed out of total mothers who agreed to take part in maternity unit	100.0	84.4	75.4	69.9
Apparent cumulative attrition rate[b] of mothers who agreed to take part in maternity unit (%)	0.0	15.6	24.6	30.1

[a] These births occurred in maternity units where the pilot survey was held: 915 births were eligible in principle, but approximately 9% of the women (81) who agreed to participate in Elfe while at the maternity unit were withdrawn from the sample (ineligible) for the second two phases because they did not live in the pilot *départements* at the time of the 2-month survey.
[b] The attrition rate is "apparent" because it takes account of the fact that some non-participants in one wave responded to a later wave.
Sources: Elfe pilot surveys, 2007.

higher in level 1 maternity units than in those with a neonatology unit or those with a neonatal intensive-care unit. It was also higher in the Seine-Saint-Denis *département* than in the others (Sarter et al. 2008).

Causes of failure were varied, and in some cases liable to generate recruitment bias. They were mainly due to midwives not having time to present the survey, women refusing to take part and the difficulty of conducting the survey with particular groups, such as people who do not speak French or mothers of newborns urgently transferred to another hospital ward. Midwives were paid for this work in the maternity unit but many of them combined their daily routine with the administration of the survey. Where this happened, the midwives did not always have time to interview all the mothers who had just given birth. Some refusals were doubtless also due to the survey being presented too rapidly, without giving mothers sufficient time to understand the purpose of the study. Some women expressed distrust of statistical surveys, others wanted to rest or preferred to spend the time with their new baby at a very emotional moment in their life. Initial analysis of the women's socio-demographic characteristics shows that the acceptance rate rises with age, occupational status and educational level. For the full-scale survey, an association of interpreters (Inter Service Migrants) will help out in the maternity units so that language is not an obstacle to participating in Elfe.

The second phase of the pilot survey was held at age two months. In an interview lasting 60 minutes, mothers were asked face-to-face about various

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topics: the family's socio-demographic characteristics, the child's health, relations within and outside the family, education, type of childcare and diet. A 35-minute telephone conversation on the same topics was held with the father. These interviews were carried out by INSEE. The interviewers also left a nutrition diary and either a radon sensor or a dust trap at the children's homes.(7) The participation rate in the two-month survey was satisfactory, at 85% for mothers and 79% for fathers among the families who had agreed in the maternity unit to take part in the study and were still living in the *départements* chosen for the pilot study. Of those who did not respond, approximately 5% could not be contacted, either because they had moved or because we did not have sufficiently accurate postal addresses or telephone numbers. During this phase, the families expressed their interest in the Elfe study and the various topics they were asked about, particularly those specifically relating to child development and health, and relations between parents and children. However, many seemed still not to have fully understood the study objectives. The interviewers observed that mothers and fathers differed in their willingness to take part: whereas the mothers' reactions and responses were generally positive, the fathers sometimes seemed reluctant to participate and doubted that the study was anonymous. Their refusal was probably due mainly to a lack of information about the survey. Overall, it emerged that longitudinal monitoring is a very multifaceted process, requiring a thorough knowledge on the part of interviewers of all the stages and forms of observation – which will inevitably grow in complexity over the years. For example, during this second phase, many interviewers pointed out that they did not know enough about the way the survey was carried out in the maternity units to answer some of the questions women asked. We shall need to find various formats (flow charts, information brochures, etc.) summarizing the stages that precede the interview phase in order to make the study presentation more transparent and explicit overall. For the subsequent pilot surveys (at 2 and 3 years), non-response will be monitored more closely (self-completion questionnaire for interviewers during field work) to better understand the reasons for attrition.

The survey on or near the children's first birthday was the third stage of the pilot study. It was held by telephone and mainly concerned the topics already raised in the two-month survey. To test this phase, the two pilot cohorts of April and October 2007 were combined to obtain a larger sample and facilitate data collection, so some children were one year old and others 18 months at the time of the telephone survey in October 2008. In the 419 eligible families, 343 mothers took part, a participation rate of 82%, significantly higher than for fathers (70%). Out of all the families, 6% could not be interviewed because of incorrect contact details or no contact person, and 1.7% because of a language

(7) The radon sensor measures the concentration of radon in an inhabited room. The dust trap measures children's exposure to dust and mould inside the home. These measuring devices are placed in the room where the child spends most time.

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barrier. Of the families that did not want to take part in the one-year survey, 25% had already refused the at-home survey at two months.

The results of the one-year pilot survey were satisfactory in terms of feasibility and acceptability to families, but some of the problems raised in the first wave (maternity unit and two months) have not yet been resolved. For subsequent stages, thought will be given mainly to involving father's more closely in the study, developing the loyalty of families via regular contacts, encouraging long-term commitment, and providing translation support for families with difficulties understanding French.

In the Elfe pilot cohort, the proportion of (eligible) births that were not recruited was relatively high, but once recruitment was achieved, the participation rates were very good: among the women who agreed to the pilot study and were contacted after the first survey wave, 15% did not take part in the two-month survey and 18% in the one-year and two-year surveys. Experience with child cohorts in other countries has shown that lower attrition after the first survey is frequent in longitudinal studies; in general, by that stage, the respondents have fully grasped the concept of the study and wish to take part, even over the long term. A satisfaction survey was carried out among Elfe mothers and fathers after the second survey wave (age 2 months). It was done by internet with those mothers and fathers who had given an e-mail address when they agreed to take part in the Elfe pilot study. Of those asked, 47% of mothers and 28% of fathers completed and returned the online satisfaction questionnaire. Overall, the respondents are willing to take part on a long-term basis. They felt they had been properly informed in the maternity unit (88% of mothers and 77% of fathers felt well-informed about the Elfe study), although the objectives of the longitudinal analysis were not always clearly understood (74% of mothers and 82% of fathers stated that they had "not really understood" or "not understood at all" the Elfe objectives).

Combining the rates observed for each wave, starting with the maternity unit survey, a cumulative attrition rate is obtained that is apparent only, because this type of calculation ignores the fact that some non-respondents in one wave may have taken part in a later wave. For example, out of the 390 families contacted at age 2, a total of 318 responses were obtained, including 26 from families that had "missed" at least one earlier contact; similarly, of the 72 non-respondents, only 19 are considered to be definitive drop-outs.

3. Other studies and tests during the pilot phase

In order to develop the Elfe methodology, a number of specific studies were carried out.

All cohort studies are based on long-term monitoring whose acceptance by families cannot be taken for granted. For an ex-post assessment of attitudes, a qualitative study of the perceptions of people who were taking part, or had taken part, in the Gazel cohort was conducted. Gazel is a cohort of 20,000 current and

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retired employees of the power utilities EDF and GDF (three-quarters men), designed to monitor their state of health and major risk factors via epidemiological studies. The cohort began in 1989 and is still active. At present, 16,000 respondents are taking part. The qualitative study was carried out by Géraldine Vivier (INED surveys department) with some twenty respondents. It showed that they had incomplete knowledge of the project, both in terms of the information they were initially given about the collection methods (active and passive) and their right to agree or refuse to participate, and about the functioning and purpose of the cohort. The survey also revealed the need for ongoing guidance, with basic information repeated throughout the study period, because the respondents are generally rather distrustful of health studies.

A validation study was conducted to test the collection of information on eating habits during the last trimester of pregnancy (diet-nutrition group). This retrospective data collection method clearly raises problems of reliability, since respondents have to think back several months. A double protocol was tested on a sample of pregnant women: first, a 24-hour recall (R24) questionnaire to elicit details of everything the respondents had eaten in the past 24 hours (administered three times between weeks 27 and 35), a method that is generally considered to be fairly reliable; and second, their responses to the Elfe retrospective questionnaire (RQ) in the two weeks following the birth. Initial results show that the RQ gives a significant overestimate of food consumed compared with R24, but that the two methods provide quite similar results for ranking respondents by their consumption of various nutrients. For studies of the relative effects of nutrient intakes, the latter criterion is more important, so the RQ technique can therefore be used (Ahluwalia et al., 2010).

Another test examined the possibility of administering by telephone a module given at age 2, validated in a face-to-face interview, concerning early language development (psychomotor development group). This is a short version, adapted to the French context, of the MacArthur-Bates Communicative Development Inventory (CDI) to test children's knowledge of a hundred or so words. The questionnaire is usually completed by the mother at the interviewer's request. Since the Elfe two-year contact interview is to be conducted by telephone, we needed to check that the list can be read out by the interviewer and properly understood by the mother. The result is positive, except for one or two words that may be ambiguous when spoken.

The second part of the pilot survey, which involved biological sample collection, served to test the feasibility of a set of procedures that had never previously been applied to large samples. It led us to modify the protocol for collecting umbilical cord blood, and enabled us to assess the constraints of large-scale collection from a large number of hospitals. The pilot study also revealed an abnormal concentration of phthalates in certain samples, which may have been contaminated by the equipment used for sample collection (Vandentorren et al., 2011).

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4. Project ethics and information given to respondents

The collection of data and information in a project of this scale calls for very careful precautions to guarantee the respondents' anonymity and to inform the families of the study objectives. In other countries, this type of project is generally overseen by an ethics committee (university-based in most cases). In France, it must be approved by the personal data protection authorities (CNIS and CNIL) and, for biological samples and health examinations, by a supervisory committee for biomedical research involving human subjects (CPP*) and by the CCTIRS*, a consultative committee for data processing in health research.

Anonymizing and securing the data

Very close attention has been paid to the issue of data protection and anonymization. By its very nature, a cohort study involves the storage and updating of the names, addresses and telephone numbers of families in the sample. For this reason, data cannot be totally anonymized after each survey wave. Techniques are now available for updating the database by adding newly collected data for individuals while preventing any return from the database to individuals' identifiers. This type of procedure presents two problems, however. First, a single large database has to be created that contains all the information collected. This was something we wished to avoid, as we did not want to give the impression that anyone who had access, legitimately or otherwise, to a file "would know everything" about the lives of the children included (assuming they could identify them). Second, this procedure makes it impossible to access information already collected which can be used to simplify interviews at a later stage, and carries the risk of an accidental but complete loss of connection with the identifier file, which is required for monitoring. Two initial principles were thus adopted: separation between the identifier and address management file and the collected data file(s); and no creation of a single database containing all the information collected. The solution considered was a "trusted third party" (TTP), that would possess the keys for moving between the identifier file and the databases, and possess that information alone. In the end, a much more sophisticated variant of this technique was developed, the first to be implemented for a project of this type.

This approach is based on a number of general principles. The information security constraints are applied at four levels: level of use, level of technology, level of threat and level of regulation. Two information systems were designed, one for data storage and retrieval, which will exist for the entire duration of the study, and the other for data collection, which will be specific to each survey. To manage risks and information security in general, we set up an information security management system that complied with the ISO 2700x family of standards, with the objective of achieving ISO 27001 certification for the data storage and retrieval system.

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Elfe information system rules

The purpose of the information system is to store and retrieve for the research teams the data collected in successive surveys. The following principles were adopted in system design:

- The data supplied to the research teams shall be anonymized;
- The data shall only be supplied to the research teams on the basis of a legitimate request authorized by the Elfe data access committee;
- Individuals in the cohort shall be traceable (and therefore known by name) throughout the duration of the Elfe study;
- Nowhere, even within the system, shall there be any file containing all the data on cohort individuals from more than one survey;
- There shall be no exchange or transmission of data that has not received prior authorization from a relevant committee;
- Non-legitimate data shall not be transmitted or exchanged;
- If any legitimate request for a set of data can be used to indirectly to identify an individual (by cross-reference), authorization shall not be granted;
- A research team shall not be able to merge at individual level any sets of data it may have obtained further to different authorized legitimate requests.

Tasks are allocated among various distinct and autonomous entities:

- The identifier management entity is responsible for generating and storing identifiers and for storing the links between data on individuals. An individual's unique identifier is generated and stored by this entity and is never transmitted or exchanged; this entity is its sole possessor.
- The data management entity is responsible for storing actual data "in bulk", unlinked either to an individual or to other data.
- The identity management entity has the same task as the data management entity except that it only stores identity-type data such as surname or date of birth. This entity exists to apply the principle of data fragmentation.
- The join/atomization entity is responsible for atomizing data for storage, and for joining data on individuals in order to supply data for which access has been requested and authorized. This entity merely executes and never stores any files or data.
- The decision entity is responsible for authorizing access to data and for authorizing exchanges between other entities on the basis of legitimate requests. This entity transmits the necessary instructions to each of the other entities along with the resources – the "keys" – to execute these instructions. In the absence of these instructions and keys, no entity can operate or exchange with the others. This entity, as such, has no access to data; it is purely organizational.

Whatever the information system used, technical and organizational security measures are deployed to meet basic criteria of confidentiality, availability, integrity and authenticity. They include: encryption and electronic signatures for data exchanges, user authentication, security of application developments, firewalls to secure networks, backups, regular updates of systems and applications, equipment redundancy (hardware, electrics), premises security (fire, flood, intrusion), third party management, and, not least, prevention, staff training and awareness (see Box).

One general principle is that the collection system will only contain the data for one survey at a time. The data collected by the interviewers will be temporarily centralized in this system to permit quality control of the survey

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data. The second principle is that once the data are validated and transferred to the storage system, they will be deleted from the collection system. The individual's unique identifiers, essential for carrying out a survey, are survey-specific and temporary, since they are different from those in the dedicated storage system entity.

The data collection information system is specific to each survey since each one is unique. However, whatever the survey procedures, solutions and tools used, the formal and functional approach requires that basic principles be applied to each of these systems, especially if they are designed or managed by third parties.

Informing the families

When the Elfe project was presented to the CNIS in June 2007, the CNIL representative requested that the families be informed in advance, i.e. during the mothers' pregnancy, and that they receive a description of the project objectives and the topics under study. The sampling design makes it impossible to precisely identify the women who will give birth on the selected days, but the CNAF offered to send an information sheet to pregnant women alongside a routine administrative letter sent out during the seventh month of pregnancy. This procedure was tested in the pilot survey. Other, broader channels of information (press, maternity unit consultation rooms, etc.) will also be used. In general, it is essential for the information to be as transparent as possible for the parents, and later for the child, as the succession of different observation procedures and the collection of a large quantity of data may be confusing for families.

The information given to the respondents needs to explain the reasons for the study and to stress the importance of taking part. The difficulty is to motivate the respondents with sound scientific arguments while remaining clearly understandable. For example, in the brochure handed out to mothers in the maternity units for the pilot surveys in 2007, the word "longitudinal", a technical term from the vocabulary of demography and statistics, was used without explanation. The information strategy must also serve to maintain respondent participation through regular contacts in the form of birthday cards, letters of thanks, postage-paid envelopes for address changes, etc. The families taking part in the pilot survey have received a newsletter *(Elfe-infos)* every 6 months since the children's birth summarizing the study objectives and presenting some of the pilot study findings, as well as New Year's and birthday cards. The children were given a book for their second birthday. Other recurrent communication formats, such as regular media coverage (radio, press, etc.) or DVDs may be envisaged, with the intention of creating a climate of trust between the Elfe team and the respondents.

The question also arises of whether to give individual feedback to families about medical findings or exposure measurements. This point was discussed

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by the Elfe cohort "ethics group" which brings together outside experts to help the project team make certain methodological choices. At present, we consider that Elfe has neither the vocation nor the resources to make accurate medical diagnoses. Moreover, the vast majority of the biological analyses are intended to study the possible effects of various exposures for which there are currently no recommended "alert thresholds": providing information that serves no practical purpose would have little point. We shall restrict ourselves to the cases where there such thresholds exist, such as blood lead level, one of the exposures measured for the Elfe cohort. Where the level is too high, it will be mandatory to inform the families concerned. Finally, to warn families about "positive" results would contradict the principle of data anonymity. Carli Lessof (2009) adopts a slightly different point of view: she believes there is a moral obligation to give feedback where medical examinations reveal, for example, a serious illness (cancer, etc.), so that the child does not suffer needlessly. However, she considers that feedback is not necessary in cases where the child carries a genetic disease but has not developed it, or where cognitive development is being tested, since the data are not necessarily reliable at the individual level. The findings might be misinterpreted by the family or bias the respondents' answers in subsequent waves. Whatever the principle adopted, it is important to explain to respondents why some analyses may lead to feedback and others not.

Obtaining consent

The mother's express consent will be obtained in the maternity unit. The father, informed via the mother, can refuse to take part. Later, the children themselves will be asked. Although children remain under their parents' authority until their majority, they must obviously be informed well beforehand and their consent must be obtained at a fairly early age. The age of the first interview with the child has not yet been decided, but it is likely to be before age 10, and their formal agreement could be obtained at age 13-14. An interesting initiative has been taken for the Growing Up in Ireland cohort: a Children's Advisory Forum has been set up to take account of children's advice and opinions concerning, for example, the information brochures sent out to families, the study logo design, and the clarity of questions included in the questionnaires (appropriate vocabulary). The Forum comprises 84 children who are not in the cohort and who meet in 12 committees in various schools across Ireland (7 children per committee). These committees are organized by the researchers involved in the study. An external spokesperson reports back on the meetings to the cohort management team. On the Growing Up in Ireland cohort website, a specific Children's Corner is reserved for the children in the study. Its content is produced in consultation with the Children's Advisory Forum.

For waves involving biological samples or medical examinations, or for access to external data, written consent will be requested from the children's

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parents. Tate and her colleagues (2006) have shown that in the UK Millennium Cohort Study most mothers accepted the principle of passive collection in the first wave. Written consent was obtained in the first survey wave to access medical data about the pregnancy and birth. However, mothers from ethnic minorities more frequently refused or showed reluctance, a fact which the authors ascribed partly to cultural factors that increase distrust of medical research.

Conclusion

Elfe is an ambitious project, but one that belongs to the mainstream of French research and that will contribute to a better understanding of children's development and outcomes. A multidisciplinary longitudinal study on this scale has never been done before in France, and constitutes a unique "adventure" that will enrich our knowledge in terms of research and survey methodology, and encourage closer cross-disciplinary collaboration. But to achieve its objectives and ensure its success and long-term future, Elfe must maintain consistency in its organization and the presentation of its goals, not only for the families taking part but also for civil society and the research community. At present, after six years of working together, the vast majority of the initial research teams and almost all the institutional "founding" partners are still involved in the project.

Since the success of Elfe depends upon its durability, modifications will no doubt be made in the years ahead. To ensure that the most appropriate choices are made at each stage, the Elfe priorities will need to be constantly restated and its objectives clearly defined. The task is to learn both from our own current and past pilot experiences and from what has been achieved and attempted throughout the world, by maintaining exchanges with the teams currently managing cohorts in other countries, in particular via Eucconet. Experience with these cohorts has revealed the major strengths and weaknesses of multidisciplinary, nationally representative child cohorts and has highlighted the questions to be addressed, such as difficulties with long-term follow-up, advantages and drawbacks of different collection methods, funding, logistics, ethical questions and dissemination of data to the research community (Pirus and Leridon, 2010).

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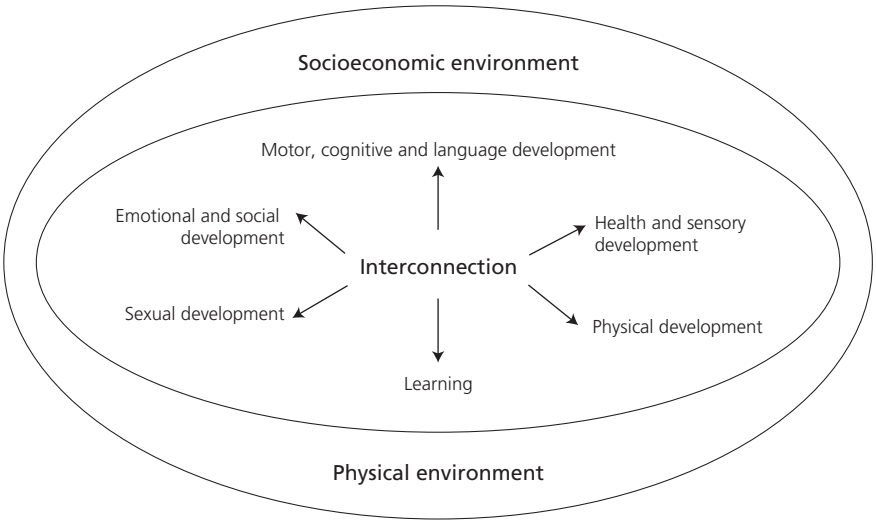
APPENDICES

Appendix 1A. The multidisciplinary dimension of the Elfe project



Appendix 1B. List of Elfe thematic groups and their coordinators (December 2010)

Social sciences groups

Demography and family: Didier Breton (Université de Strasbourg)

School: Agnès Florin (Université de Nantes)

Economic status and deprivation: Gaël De Perretti (École nationale de la statistique et de l'administration économique, ENSAE)

Socialization and education: Marion Selz (École normale supérieure, ENS)

Health groups

Accidents and injuries: Bertrand Thélot (Institut national de veille sanitaire, InVS) and Cécile Ricard (InVS)

Cancer: Jacqueline Clavel (Institut national de la santé et de la recherche médicale, INSERM)

Physical growth and puberty: Barbara Heude (INSERM)

Infectious diseases: Véronique Goulet (InVS)

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Respiratory diseases, asthma, allergies: Chantal Raherison (Centre hospitalier universitaire, CHU Bordeaux)

Paediatrics: Vincent Gadjos (Assistance publique - Hôpitaux de Paris, Bicêtre)

Perinatal period: Béatrice Blondel (INSERM)

Healthcare consumption: Corinne Bois (Protection maternelle et infantile PMI-92 and Elfe)

Mental health: Maria Melchior (INSERM) and Sylvana Côte (Université de Montréal)

Health and environment groups

Biosurveillance: Stéphanie Vandentorren (InVS)

Exposure to environmental pollutants: Céline Boudet (Institut national de l'environnement industriel et des risques, INERIS) and Christophe Declercq (InVS)

Physical exposures: Blandine Vacquier (InVS)

Cross-cutting groups

Physical activity: Patricia Dargent (INSERM and Elfe)

Diet and nutrition: Ahluwalia Naman (INSERM and Centre inter-universitaire de calcul de Toulouse, CICT) and Christine Tichit (Institut national de recherche agricole, INRA)

Psychomotor development: Catherine Arnaud (CHU Toulouse and CICT)

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Appendix 2. Organization of data collection rounds by child's age

Continuous observations	Child's age	Specific surveys
	(Pregnancy)	
Longitudinal follow-up based on the Insee Demographic Panel (EDP) Collection of data on healthcare consumption (SNIIRAM) Geocoded environmental data	Birth	Face-to-face interview (mother) + medical file + biological samples Self-administered questionnaire on nutrition and exposures during pregnancy Dust traps (dust collection during two months)
	6-8 weeks	Telephone interviews (mother and father)
	3-5 months 6-8 months 9-12 months	} Nutrition questionnaires
	1 year	Telephone interview (mother and father)
	2 years	Telephone interview (mother and father) + questionnaire filled in by family doctor
	3 years (nursery school entrance)	Face-to-face interview (mother) and telephone interview (father) + games with children + dust traps + biological samples
Monitoring of progress through school	5 years	Telephone interview (mother and father) + academic skills test
	6 years (primary school entrance) or 7 years	Face-to-face interview (mother) and telephone interview (father) + psychomotor test + medical examination and blood sample
	8 years	Telephone interview (mother and father)
	11 years (secondary school entrance)	Telephone interview (mother and father) + academic skills test (+ medical examination?)
	14 years?	Face-to-face interview

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Appendix 3. Participants in the creation of the Elfe project

The first project for a national child cohort (Cohorte nationale d'enfants, CNE) was put forward in 2002 by Henri Leridon, director of an INED-INSERM joint research unit. In June 2004, after the publication of the national health-environment plan (Plan national santé-environnement), preliminary consultations for the launch of a cohort focusing on the relationship between health and environment, known as the EFESE project, were coordinated by Georges Salines for the InVS. In June 2005, it was decided to merge CNE and EFESE, and the Elfe project was born.

The convention of the Elfe scientific interest grouping (GIS) was signed in March 2006 between the following partners: INED, INSERM, InVS, DGS and INSEE, joined by DREES, DEP and CNAF.[8] The minister responsible for sustainable development (Direction générale de l'évaluation des risques) joined the GIS at a later date. The GIS was headed by Henri Leridon until the convention came to an end in March 2010.

Claudine Pirus (INED) assisted the project director from October 2005 to October 2008, assuming particular responsibility for social science aspects. She was succeeded by Jean-Louis Lanoë (INSERM) in January 2009. Stéphanie Vandentorren (InVS) coordinated health-environment aspects from May 2006 to November 2009 and was succeeded by Ilias Kavouras in June 2010. The other health dimensions were coordinated by Corinne Bois (physician attached to the maternal and childcare services) from September 2006, who was joined by Marie-Noëlle Dufourg in March 2009. Pascal Arduin served as general secretary of the GIS from August 2008 to September 2010.

Many new members joined the team in 2009 and 2010. The GIS was replaced by a joint INED-INSERM research unit headed by Marie-Aline Charles (INSERM) in March 2010.

During the project design stage, the support of the directors of INED, INSERM and InVS was of vital importance, along with that of INSEE (notably following the report presented by Mylène Chaleix and Stéfan Lollivier to the CNIS in February 2005) and of the DGS. The support of DREES was reflected in the involvement of its senior directors and of the head of its experimentation and research mission.

The project is also backed by the ministries of research and culture, by the Mutuelle général de l'éducation nationale, by the CCDSHS and the ANR (via two calls for tenders). They have been joined more recently by the Institut de recherche en santé publique, INSERM and IRESP.

(8) See Appendix 4 for full titles and translations of acronyms.

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Appendix 4. Glossary of acronyms

Acronym	French title	Translation
ANR	Agence national de la recherche	French National Research Agency
CCDSHS	Comité consultatif pour les données en sciences humaines et sociales	Consultative committee for data in the humanities and social sciences
CCTIRS	Comité consultatif sur le traitement de l'information en matière de recherche dans le domaine de la santé	Consultative Committee for data processing in health research
CERC	Conseil de l'emploi, des revenus et de la cohésion sociale	Council for employment, income and social cohesion
CNAF	Caisse nationale des allocations familiales	National family benefits fund
CNIS	Conseil national de l'information statistique	National council for statistical information
CPP	Comité pour la protection des personnes dans la recherche biomédicale	Supervisory committee for biomedical research involving human subjects
DEP	Direction de l'évaluation et de la prospective	Evaluation and perspective directorate of the Ministry of Education
DGS	Direction générale de la santé	Health department, Ministry of Health
DREES	Direction de la recherche, des études, de l'évaluation et des statistiques	Department for research, studies and statistical evaluation at the health and employment ministries
EFS	Établissement français du sang	French blood management authority
INED	Institut national d'études démographiques	French National Institute for Demographic Studies
INSEE	Institut national de la statistique et des études économiques	National Institute for Statistics and Economic Studies
INSERM	Institut national de la santé et de la recherche médicale	National Institute for Health and Medical Research
InVS	Institut de veille sanitaire	French Institute for Public Health Surveillance
IRESP	Institut de recherche en santé publique	Institute for public health research

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Claudine Pirus, Corinne Bois, Marie-Noëlle Dufourg, Jean-Louis Lanoë, Stéphanie Vandentorren, Henri Leridon, the Elfe team • **Constructing a Cohort: Experience with the French Elfe Project**

The project to construct a multidisciplinary study cohort of 20,000 children to be monitored from birth to adulthood was launched in France in 2005. It was designed to address the concerns of various public bodies, especially those concerned with the environment, and researchers from a variety of disciplines. The result was the formation of the Elfe (Étude longitudinale française depuis l'enfance) longitudinal cohort of children, scheduled to commence in March/April 2011. The authors describe the origins of the project. The approach was deliberately intended to be multidisciplinary, and a large number of research teams were associated in designing the project, suggesting research questions in three major fields: social sciences, health and the links between health and the environment. To make the most of the life course approach, observations will be frequent (annually during the early years). A major place is given to fathers, who play an increasing role in children's upbringing and socialization. The sample will be representative of births in 2011 throughout metropolitan France. The vast quantity of information to be collected and its highly personal nature require particular caution in the creation and management of data files, for which an innovative procedure will be used. The results of a pilot cohort study launched in 2007 are also presented. They cover several hundred families, whose children are now three years old.

Claudine Pirus, Corinne Bois, Marie-Noëlle Dufourg, Jean-Louis Lanoë, Stéphanie Vandentorren, Henri Leridon, the Elfe team • **La construction d'une cohorte : l'expérience du projet français Elfe**

Le projet de mise en place d'une cohorte pluridisciplinaire de 20 000 enfants suivis depuis la naissance jusqu'à l'âge adulte a été lancé en France en 2005. Il résulte de préoccupations de diverses instances publiques – notamment dans le domaine de l'environnement – comme de chercheurs de disciplines variées. L'aboutissement en a été la constitution de la cohorte Elfe, Étude longitudinale française depuis l'enfance, dont le démarrage est prévu en mars/avril 2011. Les auteurs retracent la genèse du projet. La démarche s'est voulue résolument pluridisciplinaire, et un grand nombre d'équipes de recherche ont été associées à l'élaboration du projet, en proposant des questions de recherche dans trois grands domaines : sciences sociales, santé et relations entre santé et environnement. Pour bénéficier au maximum de la puissance de la *Life course approach*, les observations seront fréquentes (annuelles pendant les premières années). Une place importante est faite aux pères, qui jouent un rôle croissant dans l'éducation et la socialisation des enfants. L'échantillon sera représentatif des naissances de l'année 2011 sur l'ensemble du territoire de France métropolitaine. La grande quantité d'informations collectées et leur caractère très personnel exigent une prudence particulière dans la constitution et la gestion des fichiers pour laquelle une procédure originale est mise en œuvre. Les résultats du suivi d'une cohorte pilote lancée en 2007 sont aussi présentés. Ils concernent plusieurs centaines de familles, dont les enfants ont maintenant 3 ans.

Claudine Pirus, Corinne Bois, Marie-Noëlle Dufourg, Jean-Louis Lanoë, Stéphanie Vandentorren, Henri Leridon, the Elfe team • **La construcción de una cohorte : la experiencia del proyecto francés Elfe**

En 2005, Francia lanzó un proyecto de cohorte pluridisciplinario comprendiendo 20 000 niños, con un seguimiento desde el nacimiento hasta la edad adulta. Este proyecto fue el fruto de preocupaciones de diversas instancias públicas – particularmente en el sector del medio ambiente – así como de investigadores de varias disciplinas. El resultado de ello ha sido la constitución de la cohorte Elfe, Estudio longitudinal francés desde la infancia, cuyo comienzo está previsto para marzo/abril de 2011. Los autores de este artículo describen la génesis del proyecto. El proceso ha sido voluntariamente pluridisciplinario, y un gran número de equipos de investigación han sido asociados a su elaboración, proponiendo objetivos en tres grandes campos de investigación : ciencias sociales, salud y relaciones entre salud y medio ambiente. Para beneficiar al máximo de las ventajas de la *Life course approach*, las observaciones se harán a un ritmo frecuente (anual durante los primeros años). Una atención particular se presta al padre, que desempeña un papel creciente en la educación y la socialización del niño. La muestra será representativa de los nacimientos del año 2011 sobre el conjunto del territorio. La gran cantidad de informaciones recogidas y su carácter personal exigen una prudencia particular en la constitución y la gestión de los ficheros, para las cuales se aplica un procedimiento original. Se presentan también aquí los resultados de una cohorte-piloto lanzada en 2007, que concierne varios centenares de familias cuyos niños tienen ahora tres años.

Keywords: cohort, children, France, life course, ethics.

Translated by Roger Depledge.

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